



Bellabeat Case Study

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Google Data Analytics Capstone



Bellabeat

is a high-tech manufacturer of health-focused products for women. Bellabeat is a successful small company, but they have the potential to become a larger player in the global smart device market.

Bellabeat app: The Bellabeat app provides users with health data related to their activity, sleep, stress, menstrual cycle, and mindfulness habits. This data can help users better understand their current habits and make healthy decisions. The Bellabeat app connects to their line of smart wellness products.

6 Phases of a Data Analysis Project



Questions for analysis

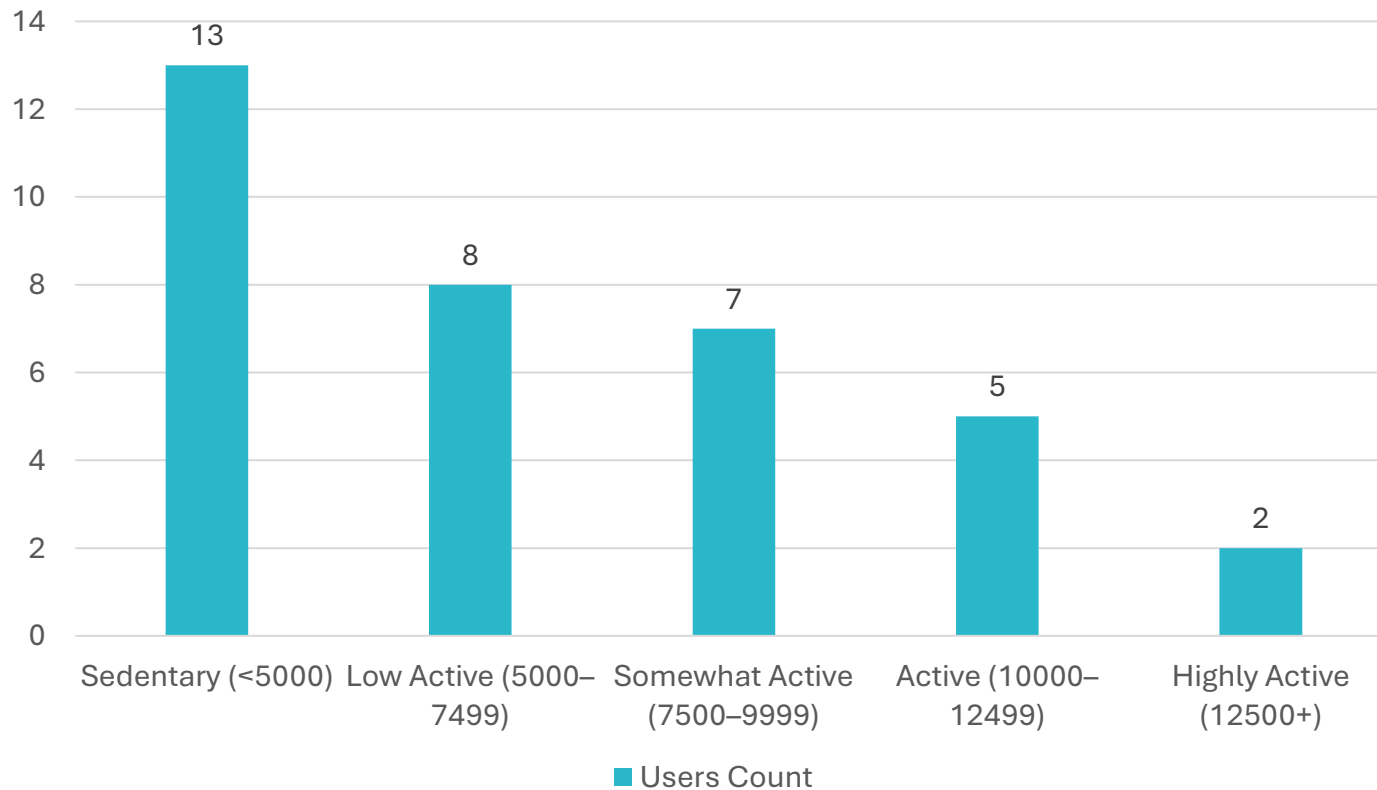
- ❓ What are some trends in smart device usage?
- ❓ How could these trends apply to Bellabeat customers?
- ❓ How could these trends help influence Bellabeat marketing strategy?

From Preparation to Exploration: Understanding User Behavior



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- After completing the six phases of data preparation and cleaning, we can move on to the most important part — understanding how users actually behave.
 - To do this, we **analyze daily and hourly activity, sleep quality and duration, stress levels, and explore correlations** between these factors. Visualizations help reveal structure and hidden patterns that shape the user experience.

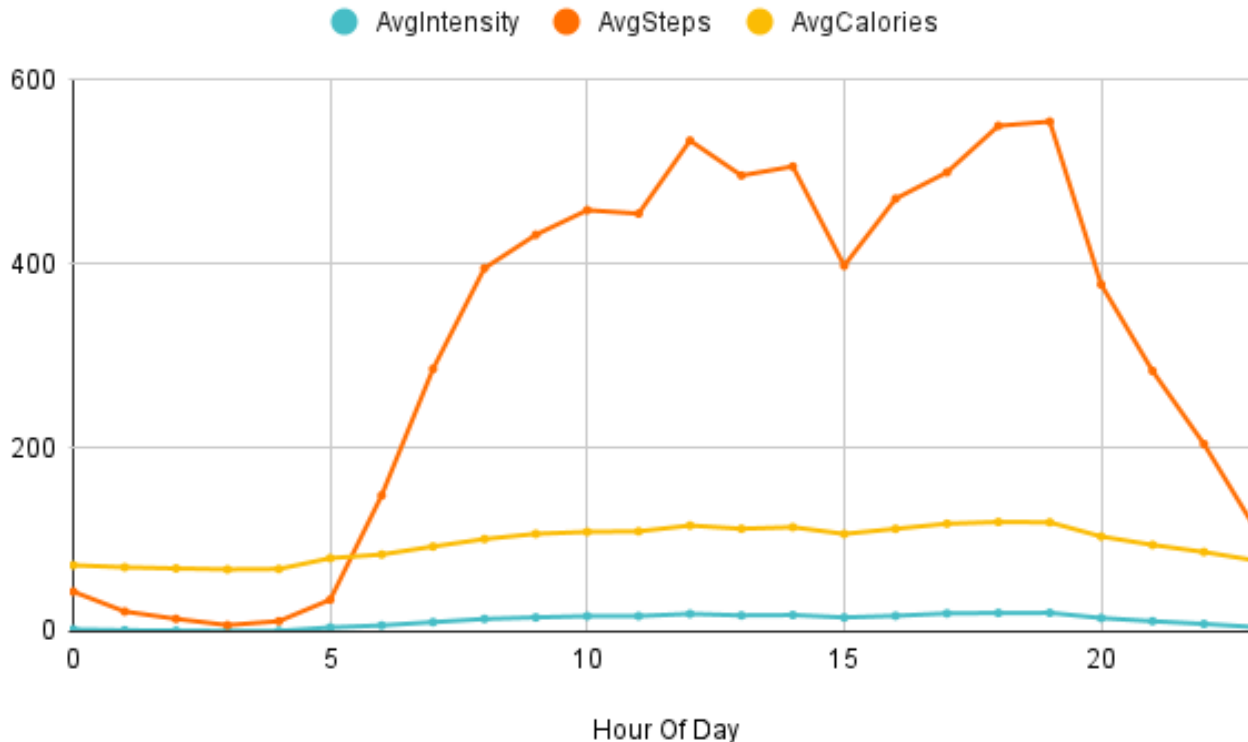
Classifying Users by Activity Level



- The largest group (13 users) falls into the Sedentary
- The next largest groups are Somewhat Active (8 users) and Low Active (7 users). Together, these categories represent moderate but inconsistent activity patterns.

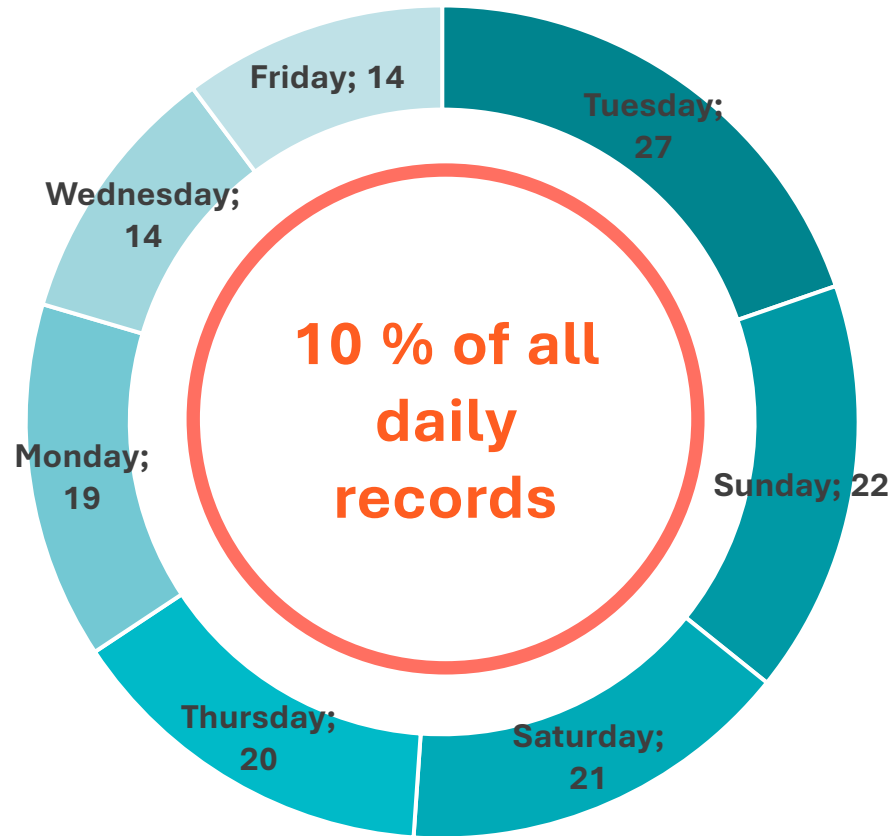
The majority of users fall below recommended daily activity levels, highlighting an opportunity for behavioral nudges, reminders, or gamified challenges.

Combined hourly analysis (intensity + steps + calories)



- Activity levels rise sharply in the morning, peak between **12:00** and **19:00**, and gradually decline toward the evening.
- Steps, intensity, and calories all follow the same pattern: very low during the night, steadily increasing after 6:00, and reaching their highest values in the afternoon and early evening.
- This reflects a **typical daily rhythm** where users are most active during work and daytime hours, with minimal movement overnight.

Identifying Days with Zero Activity



Nearly **10% of all daily records** contain zero activity across all metrics.

This strongly suggests that the device was **not used, not worn, or out of battery** on those days.

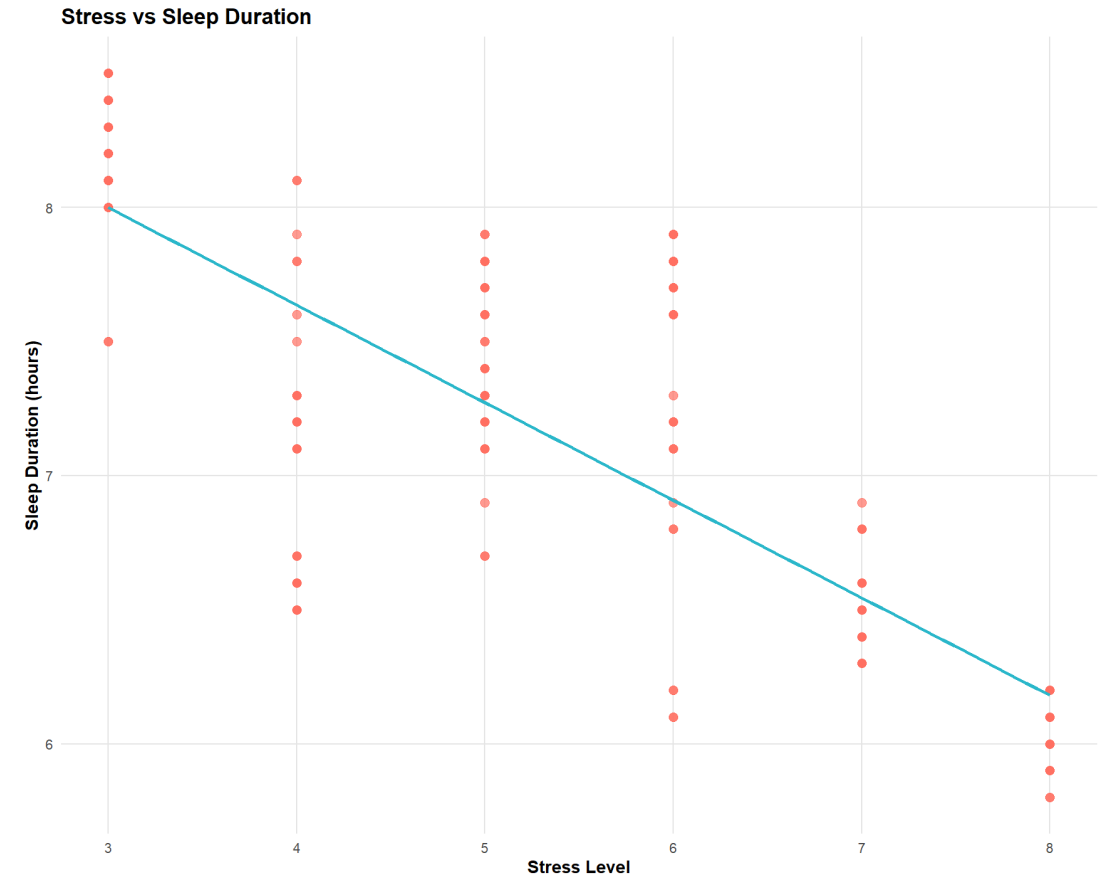
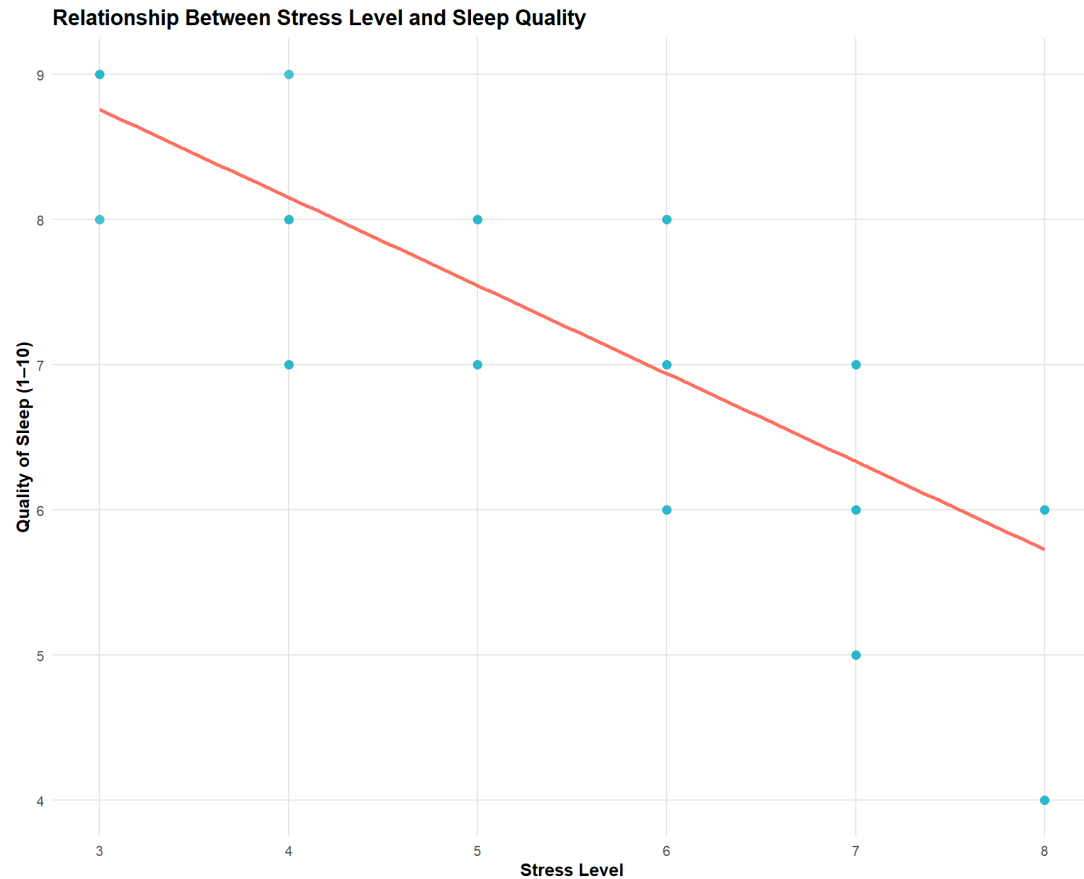
Sleep Analysis

- Sleep quality is a crucial component of overall well-being, and understanding how users sleep can **help identify behavioral patterns**, device usage habits, and potential areas for improvement.

Correlation Matrix of Key Numeric Variables

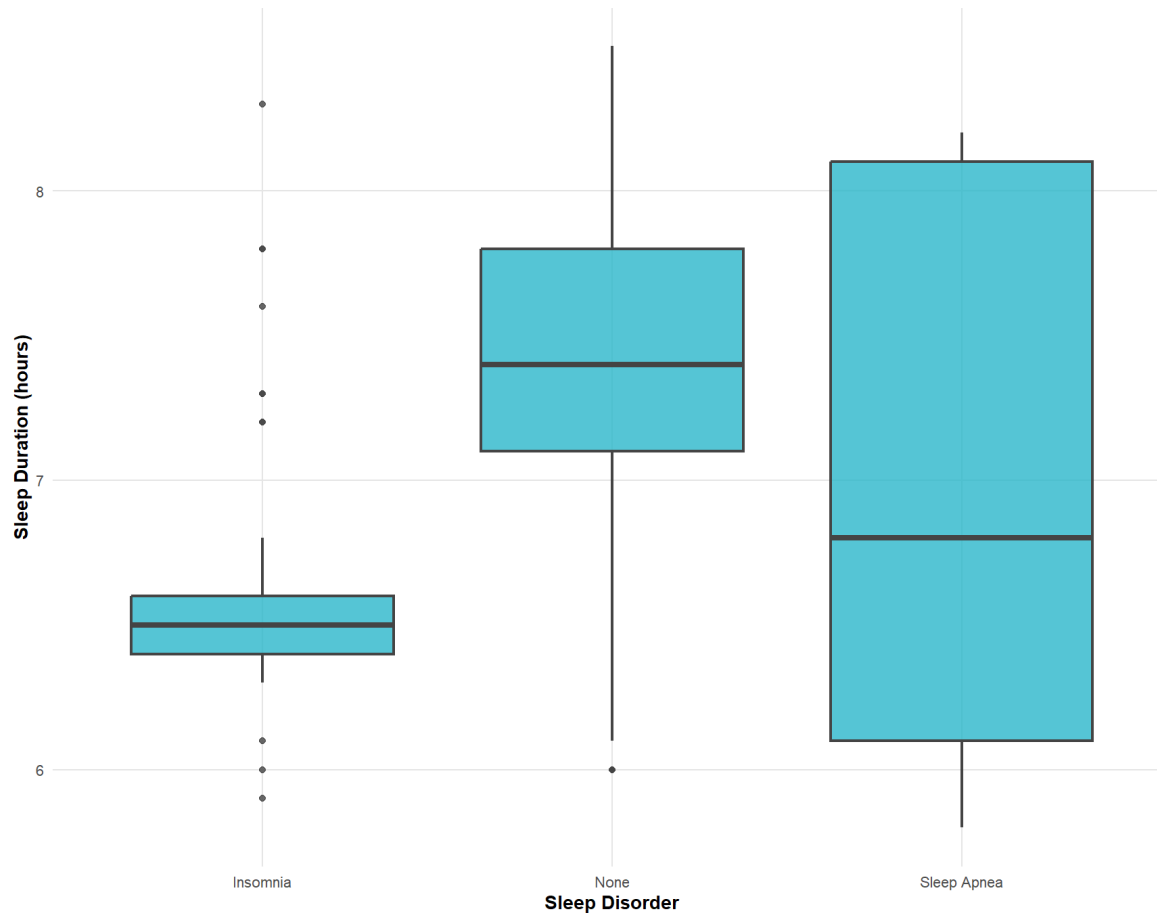


The relationship between stress and sleep indicators



- The negative slope indicates that higher stress levels are associated with lower **sleep quality and sleep duration**, which aligns with expected behavioral patterns.

Sleep Duration by Sleep Disorder



- Participants **without** a diagnosed **sleep disorder** tend to show higher sleep duration and **more stable patterns**.
- Groups with Insomnia or Sleep Apnea show **lower median sleep duration** and greater variability, which aligns with known clinical patterns.
- The stress levels in these groups also tend to be higher, reinforcing the connection between stress, sleep quality, and sleep duration.

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What are some trends in smart device usage?

- **Personalized insights** and actionable recommendations are becoming more important than raw data.
- Smart devices are shifting toward **holistic wellness**, not just fitness metrics
- Many users struggle with consistent device usage — **forgetting to charge**, removing devices at night, or skipping wear on weekends

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? How could these trends apply to Bellabeat customers?

- Bellabeat users are likely to value stress, sleep, and recovery insights as much as activity tracking.
- They benefit from clear, **personalized guidance** that connects lifestyle habits with well-being
- Usage inconsistency suggests an opportunity for **gentle reminders**, charging prompts, and habit-building nudges.

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 **How could these trends help influence Bellabeat marketing strategy?**

- Position Bellabeat as a **holistic wellness companion**, not just a fitness tracker.
- Highlight features that support **stress management, sleep improvement, and daily balance**.
- Emphasize personalization: **“Bellabeat understands your lifestyle and helps you build healthier habits.”**
- Promote **usage-boosting features** (charging reminders, wear-consistency insights) as part of a supportive ecosystem.